WEEK 3

patch  → regenerated by 

(isopotential cell)

inside cell (potential)  sending + current inside

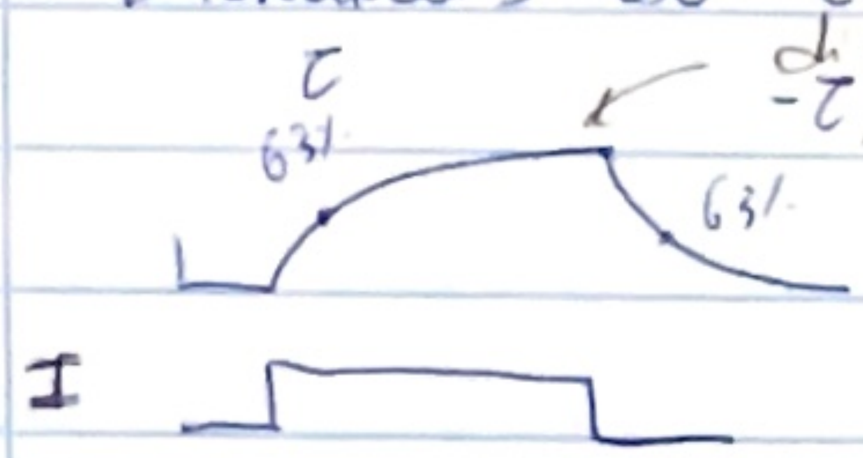
because it behaves as a RC Circuit ↑

this cell can be regenerated =

BACKGROUND ON ELECTRICITY

depolarizing current

the cell is not a resistor



(Intensity)

V (Voltage) RC system

I $t \to RC$, $V = 0{,}63\,IR$

$t \to 0$, $V = 0$

$t \to \infty$, $V = IR$

outside

inside

patch of membrane

otherwise the Voltage would be linear

$V(t) = IR(1 - e^{-t/RC})$ $RC = \tau$ constant

where its

$\tau >>>$ → long time to charge / discharge

$\tau <<<$ → less time to charge / discharge

$\boxed{I = RC \quad \text{Kirchhoff law}}$

τ_m

|

membrane time constant $t = RC = \tau_m$

|

controls how the V ↑ or ↓

TEMPORAL SUMMATION

↓

it is due to the fact that we have τ constant

electrical memory IR is the maximum you can get

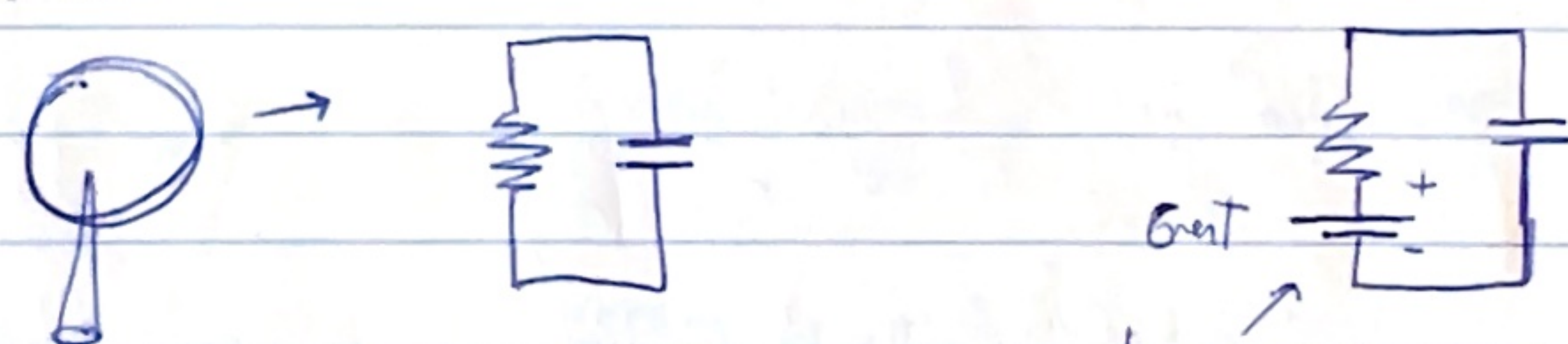
→ synaptic inputs

not too long so it summate

(in the order of τ)

τ short, V takes short

electrical elements



battery

Resting potential -70mV

I add a battery is the resting potential

τ long → V takes long

↓

tells you about the memory of the cell

hyperpolarization

depolarization

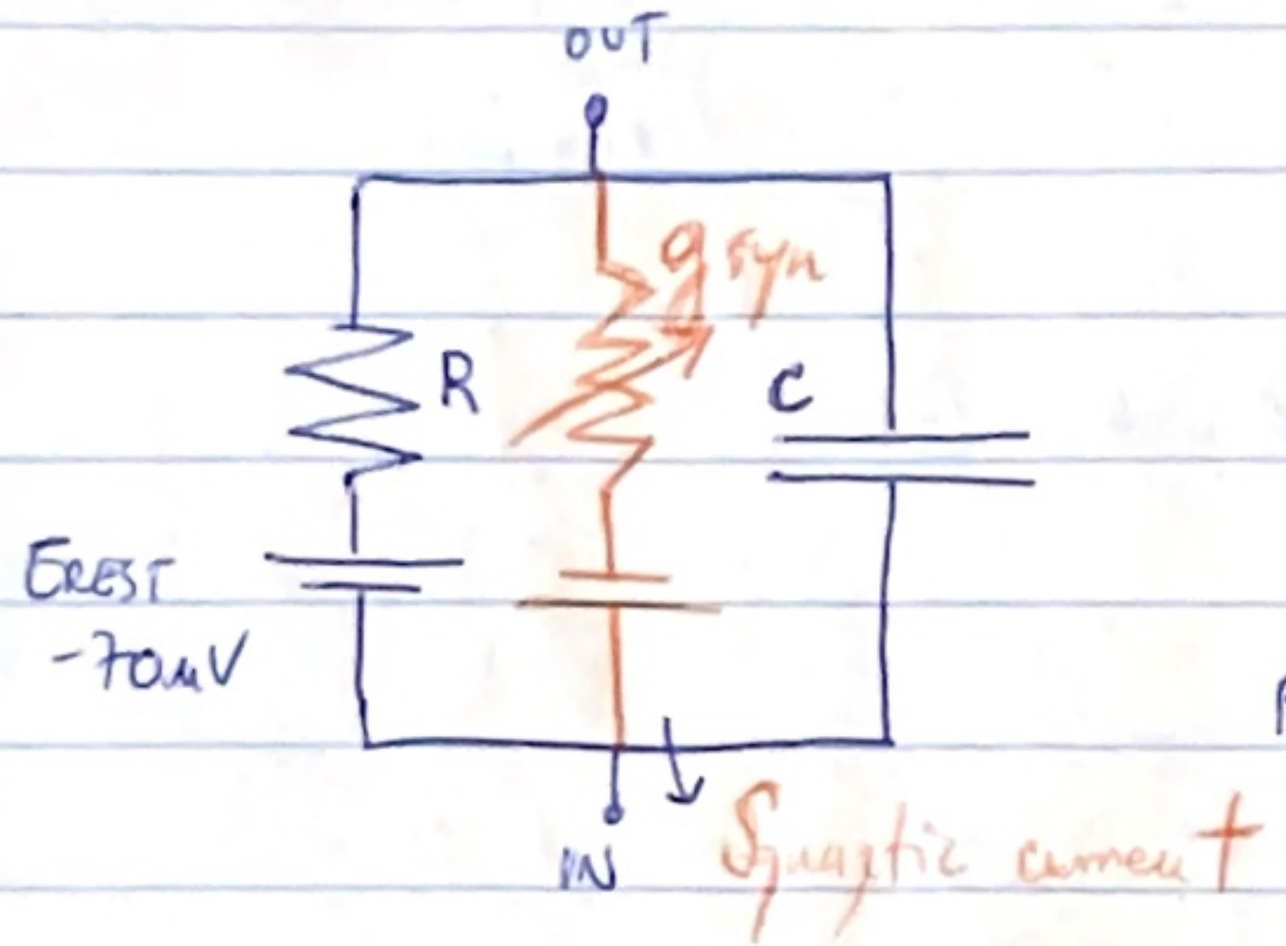
KIRHOFF

$$C\frac{dV}{dt} + \frac{V}{R} = I$$ → Injected current

capacitant current

resistive current

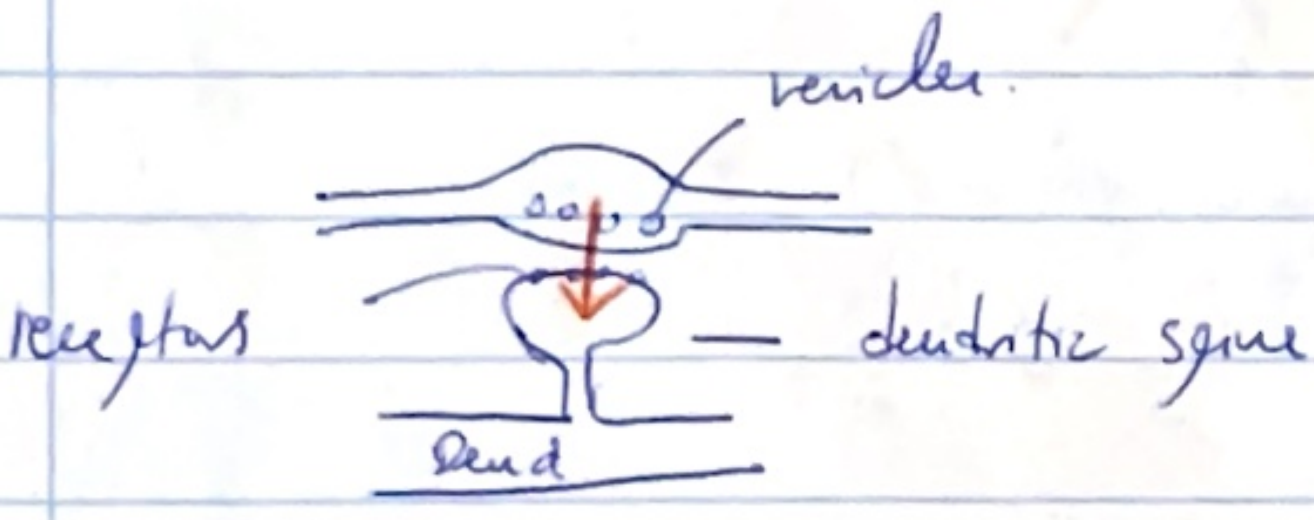
THE SYNAPTIC POTENTIAL



Neurons as passive RC circuits

Synapsis as the electrode

$R \cdot C = \tau_m \sim$ 20msec



Synapse → open ion channels
depending which one
↓
the direction of the flow

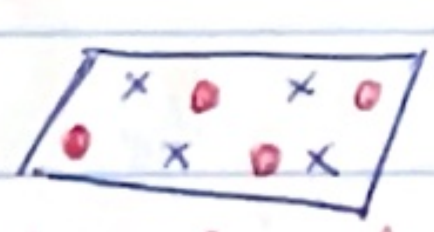
/

each channel has a conductance → Total conductance $= \frac{1}{R}$

(gamma)
|
don't depend on anything

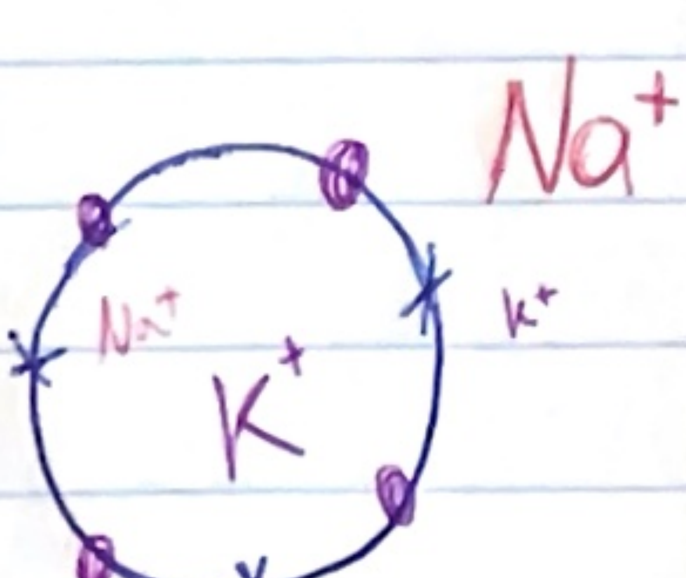
$g_r + g_r$

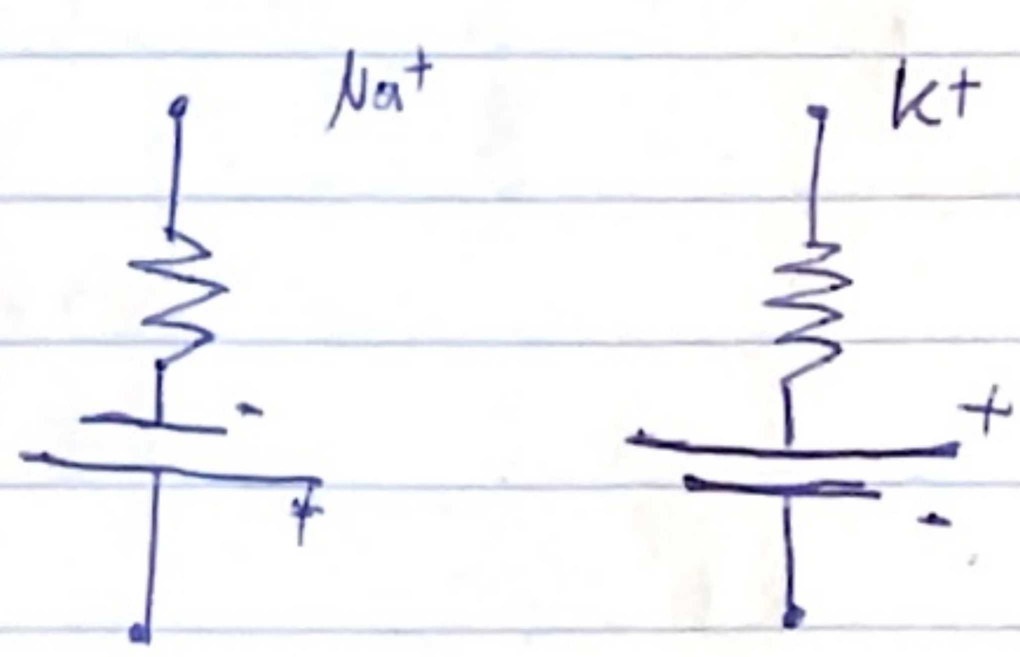
ductancia



Synaptic channels → Ion channels + $g_{synapse}$
open if transmitters are being released

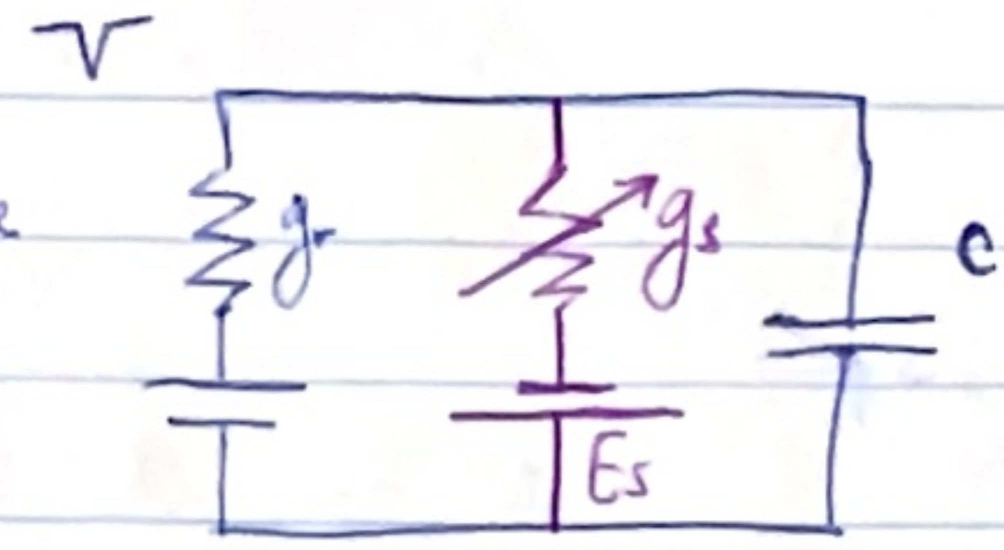
THE SYNAPTIC BATTERY





PSP

g = conductance



$$0 = \overbrace{C\frac{dV}{dt}}^{\text{capacitance current}} + \underbrace{g_r(V - E_r)}_{\text{passive current}} + g_s\cdot(V - E_{syn})$$

synaptic current

Ohm

Kirchoff's law

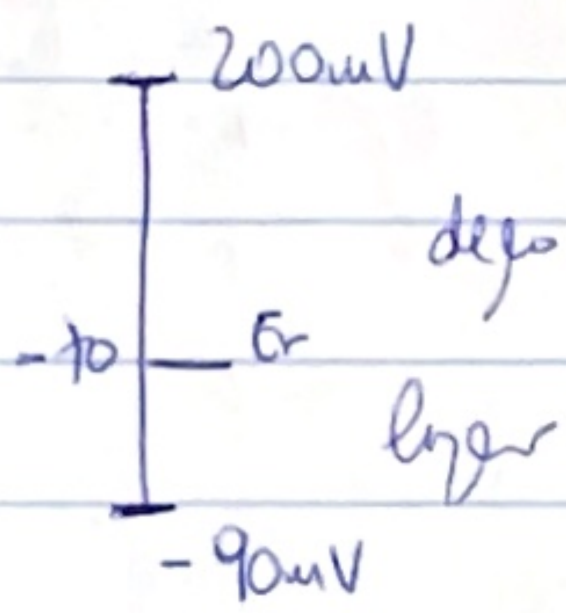
Conductance

Kirchoff's law (?)

$$V(t) = \frac{g_r E_r + g_s E_s}{g_r + g_s}$$

→ In the nervous system the Voltage changes related to batteries.
(No current from outside)

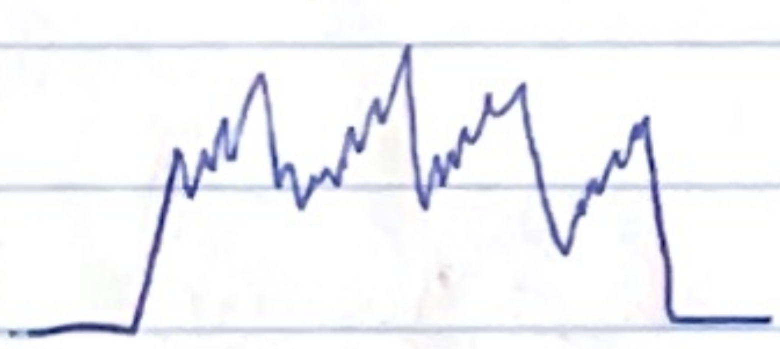
⇓



Max or minimum Voltages are the batteries.

Positive battery
↓
Na^+ (+ ion get in)

Inhibitory postsynaptic potential IPSP
excitatory " " " EPSP



temporal summation (IPSP + EPSP)